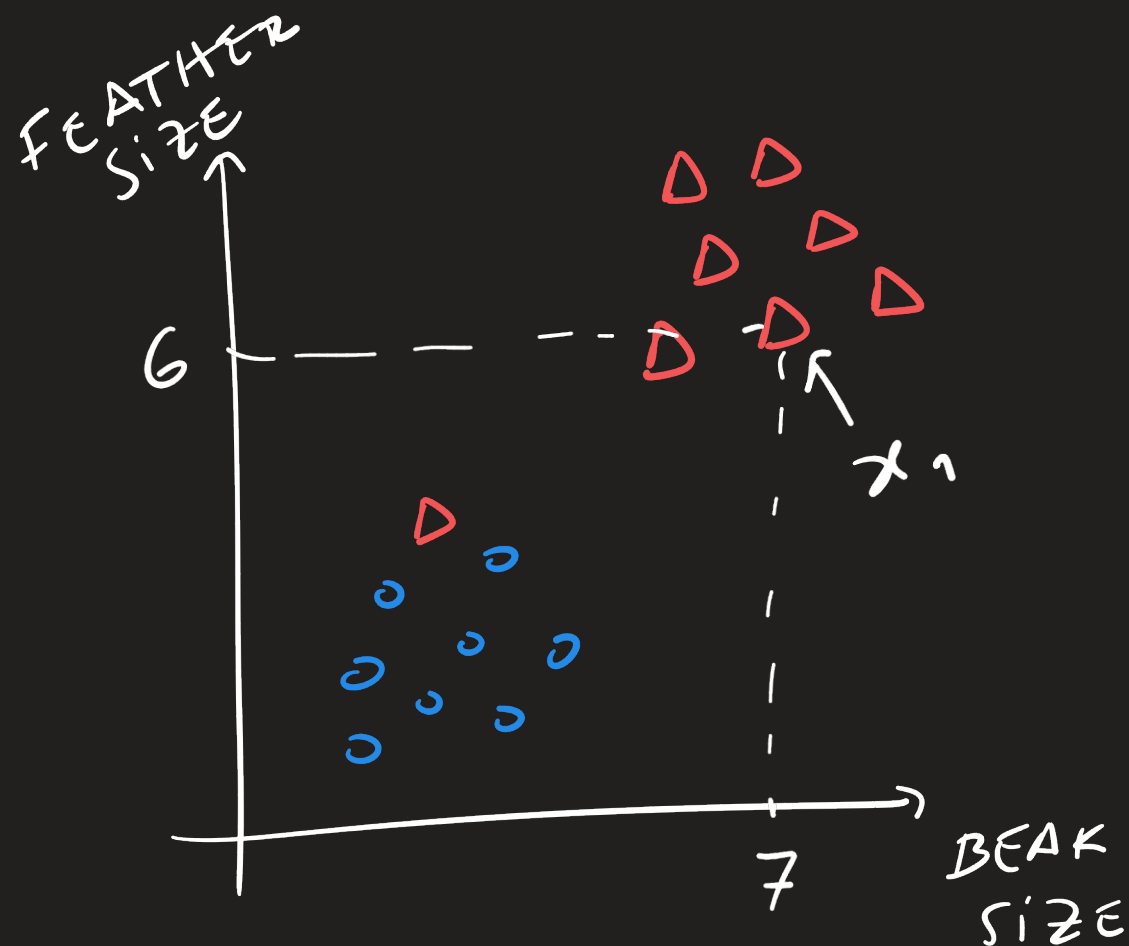


SUPERVISED Learning



FEATURE SPACE

INPUT (IMAGES)

Δ - MACAW (class 0)

$\circ$ - CONURE (class 1)

DATASET: $\{(x_i, t_i)\}_{i=1}^N$

$x_i \in \mathbb{R}^2, t_i \in \{0, 1\}$

$x_1 = [7, 6]$ input

$t_1 = 0$

Feature Matrix, X :

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

$N \times 2$

more generally

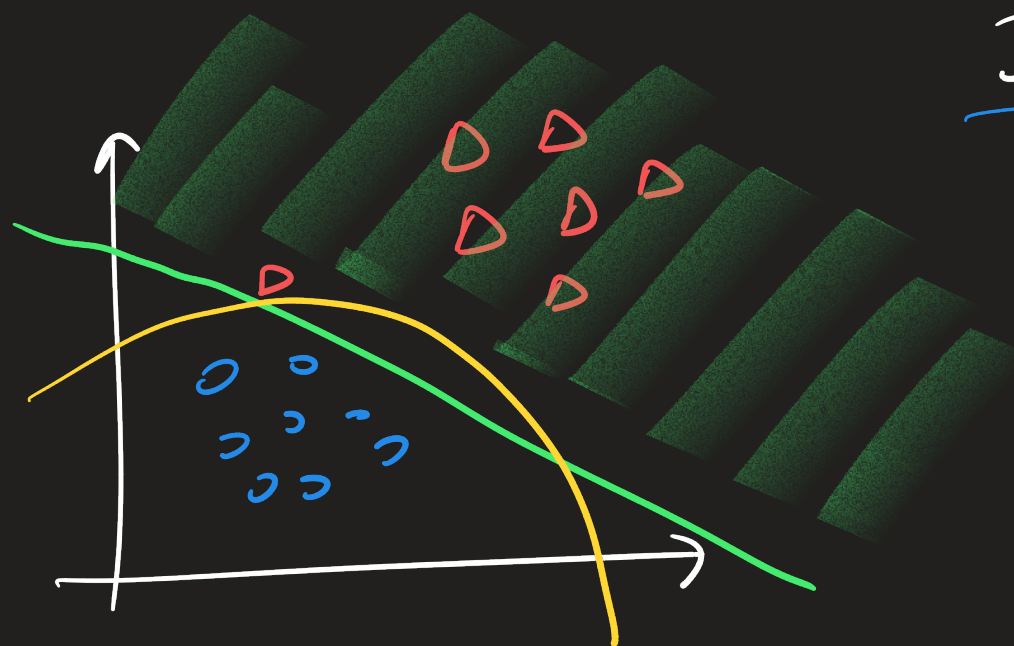
features
or
dimensions



$N \times D$

training
samples

classification



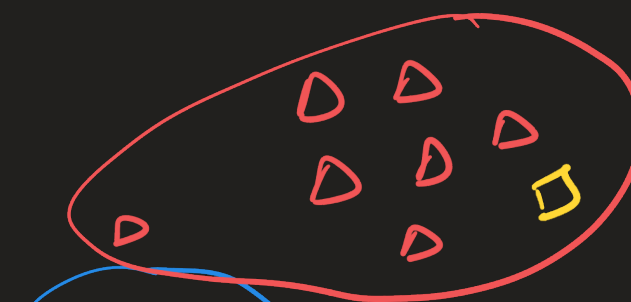
Discriminative

data likelihood

$\uparrow$
 $P(x|C_1)$

Generative

$P(C_1), P(C_0) \rightarrow$ prior prob.



$P(x|C_0 \text{ or } C_1)$

$$P(C_1 | \square) = \frac{P(\square | C_1) \cdot P(C_1)}{P(\square)} \in [0, 1]$$

$\downarrow$ Evidence prob.

Law of Total Prob.

$$P(\square) = P(\square | C_0) \times P(C_0) + P(\square | C_1) \times P(C_1)$$